Panagiotis Athanasiadis

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Personal Data

Birth: Greece, Nationality Greek
Place of residence: Thessaloniki, Greece
Languages: English (C2), Greek (Mother-tongue)

Education

Master of Science in Computer Systems — *University of Ioannina*
Department of Computer Science and Engineering — *February 2025 – Present*

Bachelor of Science in Informatics — *University of Western Macedonia*
Department of informatics — *October 2019 – January 2025*

Interests

Computational Fluid Dynamics, High Performance Computing, Numerical Analysis Software, Scientific Programming

Membership

Member at Parallel Processing Group (PPG) at the Department of Computer Science and Engineering, School of Engineering, University of Ioannina.

Internship

BETA CAE Systems — *May 2026 – November 2026*
Six-month internship in the Volume Meshing team, responsible to improve speed of computations through GPU acceleration in the ANSA software.

Relevant Coursework

Advanced subjects in Numerical Analysis — *Spring 2023*
Assistant Professor Ioannis Toulopoulos — *University of Western Macedonia*

High Performance Systems and Software — *Spring 2025*
Professor Vassilios Dimakopoulos — *University of Ioannina*

Introduction to Fluid Mechanics — *Fall 2025*
Professor Michalis Xenos — *University of Ioannina*

Thesis: Devising a scientific computing software using CAD tools — *Fall 2024*
Assistant Professor Ioannis Toulopoulos — *University of Western Macedonia*

This thesis presents the implementation of numerical methods in C/C++ to approximate solutions of differential equations with boundary conditions. Initially, we introduce the concept of B-splines as approximation tools, thanks to their geometric properties. Fundamental concepts of B-spline curves are explained, such as basis functions, the knot vector, the order of the curve, control points, and the convex hull. Subsequently, we construct a system of the form Ax = b, which we solve using the Cholesky method.

Projects

High Performance CFD Solver for Laminar Pipe Flow Problem

Written in C++ and CUDA, this project aims to speed up execution times over the MATLAB implementation up to 50 times more by utilizing advanced synchronization techniques, parallel computations and GPU programming.

Construction of Raspberry Pi Cluster

Scope of this project is to explore parallel computations on an ARM architecture CPU.

Contributions to Open Source Projects

OCC Open Computational Chemistry is a next-generation quantum chemistry and crystallography program and library, designed for modern computational workflows, written in pure C++, my part of contribution is to improve build times by refactoring the code base and applying compiler optimizations.

Technical skills

Programming Languages C, C++, Python, LaTeX
Tools/Libraries GSL, OpenMP, MPI, CUDA, Intel Vtune Profiler, Intel Math Kernel Library, Nvidia Nsight Compute

Scholarships

Undergraduate Scholarship Compensatory scholarship by University of Western Macedonia *2023-2024*
Undergraduate Scholarship Compensatory scholarship by University of Western Macedonia *2022-2023*

Additional Training

Summer School CINI-High-Performance Computing for Science, Industry, and Society *June 2025*
Summer School University of Thessaly–Dynamic Systems and Complexity *July 2025*